

1. Strategy

Let $K \subset S^3$ be a nontrivial knot, $Y_r = S_r^3(K)$, and $\phi: Y_{-2} \rightarrow Y_2$ an orientation-preserving diffeomorphism. By [DLME, Cor. 1.4], after Ni–Wu and Hanselman, this is the last open case of the cosmetic surgery conjecture, and the Chern–Simons filtration, which settles $\pm 1/n$, is unfavourable for the analogous maps, such as B below [DLME, §1.2]. We replace it by an obstruction from $\text{SO}(3)$ -monopoles. Thanks to ϕ , copies of B compose indefinitely, keeping the pairings of Step 2 nonzero, but each brings a sphere of square -4 whose μ -class costs every Seiberg–Witten stratum more energy than it supplies (Step 3), until no stratum can carry the nonzero count that Step 1 requires.

Step 1. A Witten-type relation, used in reverse. The $\text{SO}(3)$ -monopole cobordism of Pidstrigach and Tyurin, from which Feehan and Leness derive Witten’s conjecture [W] for many four-manifolds modulo a gluing theorem [PT, FLM, FL6], also gives a proved vanishing theorem. Take a spin^u structure $\mathfrak{t}$ (a spin^c structure twisted by a $\text{U}(2)$ -bundle E) with $w = c_1(E)$, $\Lambda = c_1(\mathfrak{t})$, energy $\kappa = -p_1(\mathfrak{t})/4$, and positive index $n_D = (\Lambda^2 - \sigma)/4 - \kappa$ of the Dirac operator coupled to an instanton. Near a generic instanton A the link in the quotient by the circle acting on the spinor has fibre $\mathbb{P}(\ker D_A) \cong \mathbb{CP}^{n_D-1}$, on which the class μ_c of [FL2b] is twice the hyperplane class. So for $\deg z = \dim M_\kappa^w$ the quotient cut down by $\mu_p(z)$ and $\mu_c^{n_D-1}$ is a one-manifold with $2^{n_D-1} D_X^w(z)$ ends at instantons [FL2b, Prop. 3.29]; its other ends are at Seiberg–Witten strata $M_{\mathfrak{s}} \times \text{Sym}^\ell(X)$, $c_1(\mathfrak{s}) = \Lambda + v$, of level $\ell = \kappa + v^2/4 \geq 0$, the number of bubbles. If it reaches none, Stokes’ theorem gives $D_X^w(z) = 0$ without gluing at reducibles (Theorem 2.1). In reverse: a nonzero count must reach a stratum with $M_{\mathfrak{s}} \neq \emptyset$, even if $SW(\mathfrak{s}) = 0$; we produce one where none can be reached.

Step 2. Nonvanishing. In Floer’s instanton homology I [DLME], let $B: I(Y_2) \rightarrow I(Y_{-2})$ be the map of the negative-definite cobordism $W' = (-X_2(K)) \cup_{S^3} X_{-2}(K)$ over an interval of metrics (Step 4), and $H: I(Y_{-2}) \rightarrow I(Y_2)$ that of the cobordism W_H through Y_0 . Both are defined over $\mathbb{Z}$; H is an isomorphism, and modulo two B is expected to invert it up to a fixed automorphism. So $u = (\phi_* B)^3 H B$ is invertible on $(I(Y_2; \mathbb{Z})/\text{torsion}) \otimes \mathbb{Z}_{(2)}$, and $u^M \equiv 1 \pmod{2^N}$ for some M . Caps of $\pm Y_2$ with $b^+ > 0$ and relative invariants $\langle \Psi_+, \Psi_- \rangle = q \neq 0$, obtained as in [KM] from a symplectic manifold containing Y_0 (here K must be nontrivial), give $\langle \Psi_+, u^M \Psi_- \rangle \equiv q \not\equiv 0 \pmod{2^N}$. Integrality matters: Step 1 has a factor 2^{n_D-1} , which vanishes modulo two.

Step 3. An insertion costs more energy than it supplies. The cores of the 2-handles of W' form a sphere S , $[S] = F_l - F_r$, with halves F_l, F_r from the two traces, orthogonal of square -2 ; so $S \cdot S = -4$, $\partial \nu S = L(4, 1)$, and B is cut down by $\mu(S)$. Regard κ as the energy available; a stratum spends $-v^2/4$ on its reducible and ℓ on bubbles. As $\dim M_\kappa^w = 8\kappa - 3(1 + b^+)$, a degree-two insertion supplies $1/4$, or $1/8$ with a parameter as in B . Represent $\mu(S)$ by the jumping-line divisor V_S (§2). A reducible with class v lies in V_S iff $\langle v, S \rangle \neq 0$, and then the halves, on which w , hence v , is even, cost at least $1/2$ in $-v^2/4$; if $\langle v, S \rangle = 0$, a limit lies in V_S only with a bubble on S . So a stratum is reached only if $\ell \geq T(v)$, the number of spheres orthogonal to v , and each sphere lowers $\ell - T(v)$ by at least $3/8$.

Step 4. Why a family. The closed manifold X_M capping the composite of u^M is a connected sum along the middle 3-spheres of the copies of W' , with $b^+ > 0$ on both sides, so its Donaldson and Seiberg–Witten invariants vanish. The pairing of Step 2 computes instead a secondary invariant Ω_M , a count of instantons over the cube $[-1, 1]^n$ of metrics whose faces stretch the necks S^3 and $L(4, 1)$, cut down by the $\mu(S_i)$ and the cap classes, and summed with signs over the bundles $c_e = c_0 + \sum_i e_i \text{PD}(S_i)$, $e \in \{0, 1\}^n$. So is B : summed over c and $c + \text{PD}(S)$, the map of W' cut down by $\mu(S)$ is null-homotopic by stretching S^3 , as $I(S^3) = 0$, and $L(4, 1)$, where the ends at the reducible of energy $1/4$ on νS cancel; B is the difference. The parameter also lowers the energy an insertion supplies from $1/4$ to $1/8$, and this is decisive: $\ell(K) + \ell(-K) - 2n_D = 4\kappa + (K^2 + \sigma)/2$ does not involve Λ , and a sphere with its halves changes it by $4s - 1$, s the energy supplied, for K orthogonal to the halves (the hardest to exclude): by 0 in a closed manifold, by $-1/2$ in a family. So with Λ constrained as in Step 5, a closed manifold gains nothing from spheres (Section 2).

Step 5. Why b^+ . Step 1 needs $n_D \geq 1$, else the link is empty; Feehan and Leness take Λ large [FLM, Thm. 1]. Here the relation is applied to the signed sum over the c_e , whose instanton ends carry 2^{n_D-1} , so n_D must be constant; as changing e_i changes Λ^2 by $2\Lambda \cdot S_i - 4$, $\Lambda \cdot S_i = 2$. Then $\Lambda \cdot F_l, \Lambda \cdot F_r$ are even with difference 2, the halves add nothing to $\Theta = (\Lambda^2 - \sigma)/4$, and a chain of negative-definite pieces adds at most $1/2$, from its ends: the index must come from b^+ . The positive piece $P \cong \mathbb{CP}^2 \# 5\overline{\mathbb{CP}^2}$, W_H capped by the adjacent traces, has $\Theta(P) = 1$ for the Λ its chamber condition allows, and its positive direction supplies $3/8$: P raises n_D by $5/8$. Its Seiberg–Witten classes are controlled by a sphere of square zero and by toric metrics with $\text{Scal} \geq 0$, admitting only K with $K \cdot H$ between 0 and $\Lambda \cdot H$ (H the period point); as P is indefinite, a class can pair nontrivially with both adjacent spheres at total cost $1/2$, not 1 .

Step 6. The balance. With n spheres and m positive pieces, a copy of B lowers n_D by $1/8$ and $\ell - T(v)$ by at least $3/8$; a positive piece raises n_D by $5/8$ and the bound on $\ell - T(v)$ by $7/8$, its $3/8$ plus the $1/2$ it saves its neighbours. So $n_D = (5m - n)/8 + O(1)$ and $\ell - T(v) \leq (7m - 3n)/8 + O(1)$: for large n , $n_D \geq 1$ and exclusion hold if $1/5 < m/n < 3/7$, and outside $[1/5, 3/7]$ the count fails. As limits may break along the 3-spheres, the inequality is needed on each segment of consecutive pieces; it holds once positive pieces are three copies of B apart, except next to an irreducible component with nonzero spinor, where the available argument needs four. So the average spacing is in $[4, 5]$; u realizes four. Only ϕ allows it: $-W'$ has $b^+ = 2$, so without ϕ the only return supplied by the triangles is W_H , $m = n$, and exclusion fails. It must, since Step 2 also gives nonzero pairings $\langle \Psi_+, (HB)^M \Psi_- \rangle$, with $n_D \rightarrow \infty$,

for every nontrivial knot. For u and large M , the family version of Step 1 gives $2^{n_D-1}\Omega_M = 0$, while Step 2 gives $\Omega_M \not\equiv 0 \pmod{2^N}$: the contradiction. That version is not in the literature; Section 2 proves its closed model.

2. A closed prototype

The mechanism of Steps 1 and 3 can be seen on a closed four-manifold, in a version of [FL2b, Thm. 3.33(a)] in which the insertions exclude the reducibles, though it cannot by itself give the contradiction. Let X be closed, oriented and simply connected with $b^+ \geq 1$, with generic metric g and perturbations as in [FL2b, FLM], and $\mathfrak{t}, w, \Lambda, \kappa, n_D$ (FL's n_a) as in Step 1. For a spin^c structure $\mathfrak{s}$ let $v = c_1(\mathfrak{s}) - \Lambda \equiv w \pmod{2}$, and $M_{\mathfrak{s}}$ its Seiberg–Witten moduli space, perturbed so that its points are the reducibles split by $\mathfrak{s}$ [FL2a, Lemma 3.13]; it occurs in the Uhlenbeck closure $\bar{\mathcal{M}}_{\mathfrak{t}}$ only as $M_{\mathfrak{s}} \times \text{Sym}^{\ell}(X)$, $\ell = \kappa + v^2/4$ [FL2b, Lemma 3.32, (3.64)]. Let $S_1, \dots, S_n$ be disjoint embedded spheres with $\langle w, S_i \rangle$ even, and $T(v) = \#\{i : \langle v, S_i \rangle = 0\}$.

Theorem 2.1. *Let $w \not\equiv 0 \pmod{2}$ and $n_D \geq 1$, and let z' be a monomial in the point class and classes of embedded surfaces with $\deg z' + 2n = \dim M_{\kappa}^w$. (a) If*

$$(2.1) \quad \kappa + v^2/4 < T(v) \quad \text{for every } \mathfrak{s} \text{ with } M_{\mathfrak{s}} \neq \emptyset,$$

then $D_X^w(\mu(S_1) \cdots \mu(S_n) z') = 0$ (for $b^+ = 1$, in the chamber of g). (b) Let $[S_i] = F_{l,i} - F_{r,i}$, where the $2n$ classes $F_{l,i}, F_{r,i}$ are pairwise orthogonal of square -2 and w is even on each; let π be the orthogonal projection to the complement of their span, and $c_{\perp}$ the largest $(\pi v)^2$ for $M_{\mathfrak{s}} \neq \emptyset$. If $2n > \deg z' + 3(1 + b^+) + 2c_{\perp}$, then (2.1) holds.

For $n = 0$, (2.1) is hypothesis (3.68) of [FL2b, Thm. 3.33(a)] and (a) is that theorem. The insertions enter (2.1) on both sides: each raises κ by $1/4$ and forces a bubble onto every sphere orthogonal to v .

Proof. (a) We follow the proof of [FL2b, Thm. 3.33(a)], changing only the representatives of $\mu(S_i)$; all new inputs are elementary except transversality, standard but not written out for $\text{SO}(3)$ -monopoles. As $\langle w, S_i \rangle$ is even, $\bar{\partial}_A$ makes $E_A|_{S_i} \otimes (\det E_A|_{S_i})^{-1/2}$ a holomorphic bundle $\mathcal{O}(k) \oplus \mathcal{O}(-k)$ on $S_i \cong \mathbb{P}^1$ (Grothendieck), and the coupled Dirac operator, $\bar{\partial}$ on its twist by $\mathcal{O}(-1)$, has index zero and kernel $H^0(\mathcal{O}(k-1) \oplus \mathcal{O}(-k-1))$, nonzero iff $k \geq 1$. So the canonical section of its determinant line, of first Chern class $\pm \mu(S_i)$ [D], vanishes exactly on the jumping-line divisor $V_i = \{[A] : k \geq 1\}$. (J1) V_i depends only on $A|_{S_i}$ and is closed under C^0 convergence of these restrictions, as $\bar{\partial}_A - \bar{\partial}_{A'}$ has order zero. (J2) A connection preserving a splitting $E|_{S_i} = L_1 \oplus L_2$ with $\deg L_1 = \deg L_2$ is not in V_i , as $k = 0$. A small perturbation through sections depending only on $A|_{S_i}$ gives transversality on the irreducible strata of all levels, and makes $\mathcal{V}(z)$ smooth and transverse to M_{κ}^w at the finitely many points of $\bar{\mathcal{V}}(z) \cap \bar{M}_{\kappa}^w$, as [FL2b, Lemma 3.22] requires; it keeps (J1), and (J2) near the compact set of restrictions of reducible limits with $\langle v, S_i \rangle = 0$, where the canonical section is nonzero. Put the representatives of z' and μ_c of [FL2b, Lemmas 3.12–3.13] in general position, with base points off $\bigcup S_i$. Then no point lies in supports of total codimension above four (S_i counting two), i.e. [FL2b, (3.25)]: $z = \mu(S_1) \cdots \mu(S_n) z'$ is intersection-suitable.

So $\mathcal{Z} = \mathcal{V}(z) \cap \mathcal{W}^{n_D-1} \cap \mathcal{M}_{\mathfrak{t}}^{*,0}/S^1$, where $\mathcal{V}(z), \mathcal{W}$ represent $\mu_p(z), \mu_c$ and $\mathcal{M}_{\mathfrak{t}}^{*,0}$ consists of the irreducible monopoles with nonzero spinor, is a one-manifold [FL2b, (3.26), (3.62)] with $2^{n_D-1} D_X^w(z)$ signed ends at the link $\{\|\Phi\|_{L^2}^2 = \varepsilon\}$ of the instantons: the proof of [FL2b, Prop. 3.29, (3.32)] uses only that the representatives are determined by restriction, and the smoothness above. Uhlenbeck convergence is C^∞ , after gauge, away from the bubbles [FL1, Def. 4.19, Thm. 4.20]; so by (J1) and [FL2b, Lemma 3.15(1)] a limit at level $\ell \geq 1$ satisfies every condition whose support contains no bubble, and as a level costs six dimensions and a bubble relieves at most four, the count of [FL2b, Cor. 3.18] excludes irreducible limits with nonzero spinor. There are no reducibles with zero spinor [FL2a, Prop. 3.1, Cor. 3.3]. At a limit in $M_{\mathfrak{s}} \times \text{Sym}^{\ell}(X)$, a bubble lies on each S_i with $\langle v, S_i \rangle = 0$, for otherwise the restrictions to S_i would converge in C^0 to one outside V_i , contradicting (J1) and (J2); the S_i are disjoint, so $\ell \geq T(v)$, contrary to (2.1). Hence $\bar{\mathcal{Z}} \cap \{\|\Phi\|_{L^2}^2 \geq \varepsilon\}$ is a compact oriented one-manifold with boundary on the link, so $2^{n_D-1} D_X^w(z) = 0$.

(b) The even numbers $\langle v, F_{l,i} \rangle, \langle v, F_{r,i} \rangle$ are not both zero if $\langle v, S_i \rangle \neq 0$, so $-v^2/4 = -(\pi v)^2/4 + \sum_i (\langle v, F_{l,i} \rangle^2 + \langle v, F_{r,i} \rangle^2)/8 \geq -c_{\perp}/4 + (n - T(v))/2$; with $8\kappa = 2n + \deg z' + 3(1 + b^+)$ this gives (2.1). $\square$

In (b), $c_{\perp}$ depends on Λ . If $b^+ \geq 2$ and K is basic, the moduli spaces of $\pm K$ are nonempty, so $c_{\perp} \geq (\pi K)^2 + (\pi \Lambda)^2$, and (b) with $n_D \geq 1$ needs $b_{\perp}^- > 4b^+ + 7 + \deg z' + (\pi K)^2 - 2n - \Lambda_R^2$, where $b_{\perp}^-$ counts negative directions orthogonal to the halves and Λ_R is the part of Λ along them. With Λ constrained as in Step 5, $\Lambda_R^2 \leq -2n$ and n drops out (Step 4); with $\Lambda_R = 0$ the halves add $n/2$ to the Dirac index, which the sum over bundles forbids.

References. [D] S. K. Donaldson, *Polynomial invariants for smooth four-manifolds*, Topology **29** (1990), 257–315. [DLME] A. Daemi, T. Lidman and M. Miller Eismeier, *Filtered instanton homology and cosmetic surgery*, arXiv:2410.21248. [FL1] P. M. N. Feehan and T. G. Leness, *PU(2) monopoles. I*, J. Differential Geom. **49** (1998), 265–410. [FL2a] —, *PU(2) monopoles and links of top-level Seiberg–Witten moduli spaces*, J. Reine Angew. Math. **538** (2001), 57–133. [FL2b] —, *PU(2) monopoles. II*, ibid., 135–212. [FL6] —, *Witten's conjecture for many four-manifolds of simple type*, J. Eur. Math. Soc. **17** (2015), 899–923. [FLM] —, *An SO(3)-monopole cobordism formula relating Donaldson and Seiberg–Witten invariants*, Mem. Amer. Math. Soc. **256** (2018), no. 1226. [KM] P. B. Kronheimer and T. S. Mrowka, *Witten's conjecture and Property P*, Geom. Topol. **8** (2004), 295–310. [PT] V. Ya. Pidstrigach and A. N. Tyurin, *Localisation of Donaldson invariants along Seiberg–Witten classes*, arXiv:dg-ga/9507004. [W] E. Witten, *Monopoles and four-manifolds*, Math. Res. Lett. **1** (1994), 769–796.